Applied Energy 98 (2012) 346–356

Contents lists available at SciVerse ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Relationships between meteorological variables and monthly electricity demand

Francesco Apadula *, Alessandra Bassini, Alberto Elli, Simone Scapin

Research on Energy Systems – RSE SpA, Environment and Sustainable Development Department, Via Rubattino, 54, 20134 Milan, Italy

ARTICLE INFO

Article history:
Received 7 October 2011
Received in revised form 13 March 2012
Accepted 27 March 2012
Available online 27 April 2012

Keywords:
Electricity demand
Load forecasting
Meteorological influence
Regression model

ABSTRACT

Electricity demand depends on climatic condition and the influence of weather has been widely reported in the past. The main purpose of this study is to analyse the effect of the meteorological variability on the monthly electricity demand in Italy. Temperature, wind speed, relative humidity and cloud cover are considered; the calendar effect is also taken into account. A multiple linear regression model based on calendar and weather related variables is developed to study the relationships between meteorological variables and electricity demand as well as to predict the monthly electricity demand up to 1 month ahead. The model has been extensively tested over the period 1994–2009 using different combinations of the weather related variables. Accuracies obtained are quite similar and range between 0.85% and 0.89%. Temperature turns out to be the most important variable. According to the month considered, a specific combination of the weather related variables can give the lowest Mean Absolute Percentage Error (MAPE) but differences are usually small. Good results for the summer months are obtained using Heat Index to calculate the Cooling Degree-Days; the cloud cover has a major influence from February to April.

When demand forecasts are performed using the predicted meteorological variables, an overall accuracy (MAPE) around 1.3% is obtained over the period 1994–2009.

The proposed model clearly identifies the influence of the weather conditions on the aggregated national electricity demand.

© 2012 Elsevier Ltd. All rights reserved.

1. Introduction

It is common knowledge that electricity demand essentially depends on the anthropogenic activities in the industrial, domestic and commercial sectors and, to a lesser extent, is influenced by the meteorological variability.

Models for electric load forecasting have been widely explored and formulated in the last decades at different temporal and spatial scales, as accurate load prediction is essential to the operation and planning of a utility company.

More recently, the deregulation of the electricity markets and the evidence of the ongoing climate changes [1] have promoted an increased scientific interest in the study of the relationships between energy consumption and weather/climate [2–4]. In the last decade some notable weather events have occurred in Europe (e.g. extremely high temperatures in the 2003 summer) which created a significantly greater electricity demand. As the frequency of extreme events is likely to increase in the future and relevant changes in weather variables might occur according to climate projections, it is worth to specifically examine the influence of the meteorological conditions on the electricity demand.

According to the time horizon, load forecasts can be classified into four categories: very short term forecasts (from a few minutes to 1 h ahead), short term forecasts (from 1 h to 1 week), medium forecasts (from 1 week to a year) and long-term forecasts (longer than a year). Each class differs in purpose as different time horizons are important for distinct operations within a utility company. In this study we deal with monthly load forecasting which is primarily related with the scheduling of fuel supplies, maintenance operation of the utilities and negotiation of contracts with other companies.

Differences in lead times influence the choice of the models and methods to apply as well as the selection of the important external factors affecting the electricity demand (socio-economical, meteorological, seasonal and time dependent factors). Ultimately, data availability also determines the choice of the input variables.

Various weather variables can be considered for load forecasting: temperature and humidity are the most commonly used, but wind and cloud cover are often taken into account. Composite weather variable functions, the so-called biometeorological indices, are frequently utilized to express the summer heat discomfort and the cold stress in winter that affect, respectively, the energy demand for cooling and heating.

Most electric utilities serve customers of different types such as residential, commercial, and industrial. The electric usage pattern is different for customers that belong to different classes but is

* Corresponding author. Tel.: +39 02 3992 5235; fax: +39 02 3992 4608.
E-mail addresses: Francesco.Apadula@rse-web.it (F. Apadula), Alessandra.Bassini@rse-web.it (A. Bassini), Alberto.Elli@rse-web.it (A. Elli), Simone.Scapin@rse-web.it (S. Scapin).

0306-2619/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.apenergy.2012.03.053

somewhat alike within each class [5]. Therefore most utilities distinguish load behavior on a class-by-class basis. Many authors have studied the relationships between electricity consumption and activity patterns and behavioral characteristics [6–9].

The sensitivity of electricity demand to meteorological variables is higher in the commercial and residential sectors rather than in the industrial one.

The approaches introduced for load forecasting can be grouped into two main classes: methods which apply concepts stemming from time series and regression analysis and methods which belong to the fields of Artificial and Computational Intelligence (Neural Networks, expert systems, fuzzy logic, genetic methods and Support Vector Machines). Hybrid approaches are also very common: they combine two or more different techniques in order to overcome some drawbacks of the original methods. Survey and review on models and methods can be found, for example, in [5,10–12].

In this paper a parametric multiple linear regression model has been performed to study the role of the meteorological variables on the Italian monthly electricity demand. The calendar effect and the influence of weather variables such as temperature, humidity, wind and cloud cover has been evaluated. Although it is more appropriate to perform an analysis on electricity demand data disaggregated both geographically [13] and by sectors, we used aggregated data because these were the only available one. Anyway the results obtained up to now show an evident relationship between the meteorological variability and the electricity demand. The proposed methodology could be applied to disaggregated demand data.

2. Data description

In order to analyse the relationship between the Italian electricity monthly demand and weather-related variables, a sufficiently long period of historical data is required. In this study, we have considered data covering the period from January 1981 to December 2009.

2.1. Monthly electricity demand data

The monthly demand data analyzed, made available by the Italian Company Terna, correspond to the total of all sectors of economic activity (industrial, commercial, residential, etc.) and are geographically aggregated. These data exhibit a significant and persistent upward trend until the end of 2008, mainly associated with the economic and demographic growth in the country, followed by a sudden fall in 2009 due to the global economic crisis (Fig. 1). A strong cyclic annual pattern is superimposed on this trend, mainly correlated with the changes in temperature and in the productive activities throughout the year [2,14]. The negative peaks in August clearly reflect the reduction of the productive activities during the summer holidays, while the high levels of the demand during winter (January) and the positive peaks in July are likely to be linked, respectively, to the intensive use of electric heating appliances and air conditioning systems when cold and hot temperature extremes occur. It is important to remark that the amplitude of the reported values also reflects the different length of the months and the different number of holidays and working days included in each month.

Following the approach of Moral-Carcedo and Vicéns-Otero [15] a specific variable has been introduced to capture the electricity demand variation caused by the different level of activity in the industrial, commercial and residential sectors on holidays and on working days. This new variable, which we call "weighted calendar variable", represents the effect of calendar in the demand of a particular day as a percentage of electricity demand on a representative day. To establish the calendar weights, 15 years of national daily electricity demand data were considered. For the whole daily dataset, we first computed the ratio $D_{d,w}/D_{ref,w}$, where $D_{d,w}$ is the electricity demand on a generic day d of the week w and $D_{ref,w}$ is the daily electricity demand on the reference day (Wednesday) of the same week. Second, we estimated the distribution of these ratio values for each day of the week discriminating between 'non-holidays' and holidays (mobile or fixed). The days before and after the national holidays were treated apart, considering an extended period of time in case of the Christmas and Easter holidays. The basic statistics of the ratio values assigned to each type of day was evaluated. In this way we defined the weight to assign to every type of day. The statistics, of course, is more significant for the 'non-holidays', because of the large number of events considered, and less significant for the other days.

The monthly weighted calendar variable CalW was then estimated by summing the weights of the days included in each month, both for the past and the future months. By using this variable, the role of the weather variables on the electricity demand becomes easier to identify.

To compare monthly demand values in the 1982–2009 period, the calendar effect was firstly removed standardizing demand data to a fixed-month length (30 days) and using the weighted calendar variable. The electricity demand deviations evaluated for the summer months with respect to the trend line in the period 1982–2009 are reported in Fig. 2. All demand deviations show an increasing trend along the period, with a significant growth in the last years especially for the July month, probably because of the recent widespread diffusion of air conditioning systems, boosted by the rise of tertiary activities and changes in lifestyle. The remarkable peak on June and July 2003 clearly corresponds to the exceptional summer heat wave [16] that caused several problems in Europe, both for the health and socio-economic impact [17,18].

Demand deviations corresponding to the remaining months show a slightly decreasing trend in the last decade; a more remarkable decrease occurs in April.

2.2. Meteorological variables

The meteorological dataset utilized in this work consists of the daily data acquired in 28 meteorological stations of the Italian Air Force network (CNMCA) and covers the period from 1981 to 2009. Only weather stations situated at low altitudes and close to the main towns were considered. The CNMCA dataset includes the following daily meteorological variables: temperature, relative humidity, wind speed and cloud cover.

Previous studies have shown that air temperature is the most significant climatic variable affecting electricity demand [2], whereas other variables like humidity and wind speed may be used as correcting terms for the influence of temperature [19].

The relationship between electricity demand and temperature in the European countries is non-linear with a more pronounced non-linear pattern in the warm countries [20]. An evident U-shaped relation has been found in the warmer areas, as also confirmed by other studies regarding Greece [14,21] and Spain [15,19,22].

To evaluate this relationship, we have considered a restricted area (Lombardy) in the North of Italy. Daily electricity demand data in a 15-year time interval and the corresponding daily mean temperatures, computed by means of six weather stations, were examined. The same procedure used by Moral-Carcedo and Vicéns-Otero [15] was applied to subtract trend, calendar effect and non-climatic seasonality (i.e. summer holidays) from the demand data. Once these components were filtered out, working-days demand deviations (i.e. residuals after trend and seasonality removal) were

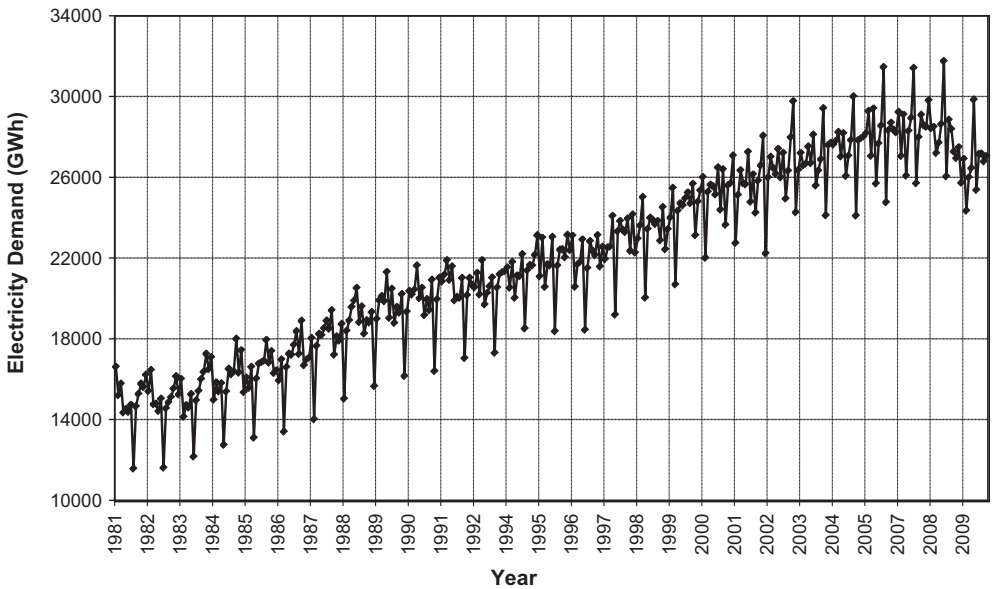

Fig. 1. Italian monthly electricity demand values. *source*: TERNA, www.terna.it.

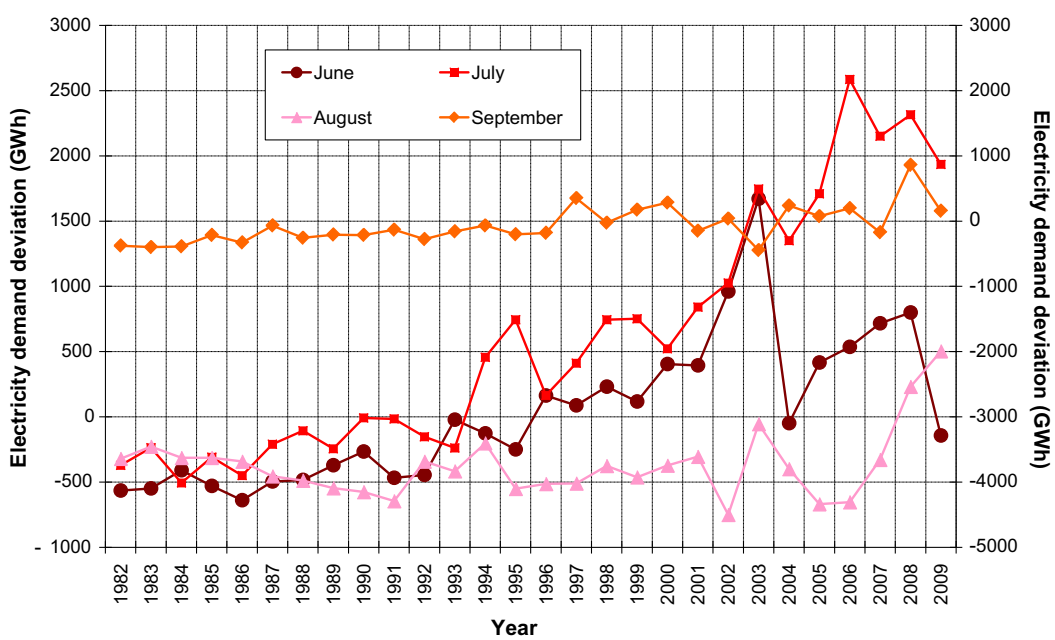

Fig. 2. Monthly electricity demand deviations with respect to the overall trend line (evaluated using a simple centerd moving average of order 12), after calendar effect removal from demand data. Summer months in the period 1982–2009 are considered: June and July values are on the left axis, August and September values are on the right scale.

plotted as a function of the mean daily temperature (Fig. 3). The scatter plot shows an asymmetrical U-shape. It is easy to see that daily mean temperatures in the range roughly between 15 °C and 20 °C do not significantly influence the electricity demand of the Lombardy region. Outside this interval, the demand considerably increases, both for low and high temperatures, with an approximately linear trend. The growth rate is stronger in the hot temperature range. Similar graphs can be obtained plotting the electricity demand deviations versus minimum or maximum daily temperatures: the U-shaped curve shifts toward lower temperatures when the minimum daily temperatures are considered and toward higher temperatures in the latter case.

In dealing with the non-linearity of the temperature effect, we have used the Cooling and Heating Degree Days variables in our linear regression model. These variables are defined in the ranges where the demand-temperature relation is approximately linear and capture, respectively, the energy requirements for cooling and heating.

In this study, the minimum and maximum daily temperatures have been used, instead of the daily average value, to effectively catch the relationship between the electricity demand and temperature; the Heating Degree Days (HDD) and Cooling Degree Days (CDD) variables were then defined in the following way:

$$\mathrm{HDD}_d = \max(T_{Ref\,H} - T_{d,min}, 0)$$

$$\mathrm{CDD}_d = \max(T_{d,max} - T_{Ref\,C}, 0)$$

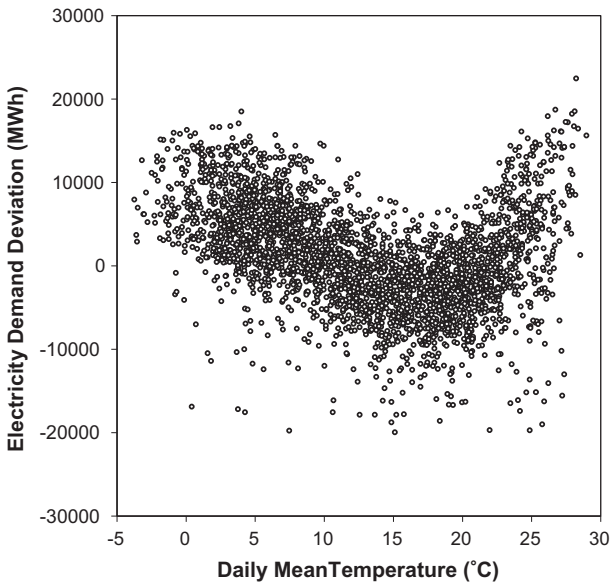

Fig. 3. Non-linearity in regional electricity demand response to temperature variations.

where $T_{d,min}$ and $T_{d,max}$ are the daily minimum and maximum temperatures for the day "*d*", $T_{Ref\ H}$ and $T_{Ref\ C}$ are properly chosen threshold values which identify the *cold* and *hot* temperature ranges respectively. The method used to identify the optimal thresholds is described in the next section.

The Heating Degree Months (HDM_m) and Cooling Degree Months (CDM_m) are then defined as the sum of all the daily values HDD_d and CDD_d included in the considered month. HDM and CDM variables represent a measure of the severity and duration of the cold and hot weather respectively.

As the available Italian electricity demand data are geographically aggregated while Italy does not have a homogeneous climate (for example, temperature often varies considerably moving from the north to the south), proper HDM_m and CDM_m values, representative of the whole peninsula, were evaluated to use in the model. The procedure adopted consists, as a first step, in dividing the Italian peninsula into a set of zones with homogeneous climate variability.[1] Secondly, HDM_m and CDM_m variables are computed in each zone using, respectively, the daily average $T_{d,min}$ and $T_{d,max}$ values of the area considered. Finally, national values of the Degree Months variables are obtained summing all the areal values.

For the determination of the zones with homogeneous climate variability, a Principal Component Analysis (PCA) with Equamax rotation was applied to the maximum and minimum daily temperature values registered in 37 meteorological stations distributed over Italy. Considering how the climatic conditions vary along the country, five principal components were searched for and detected. This preliminary grouping well matched with the geographical location of the stations, therefore five climate areas were roughly identified: North West (NW), North East (NE), High Peninsula (HP), Central Peninsula (CP) and South and Islands (SI).

The climatic variability of the meteorological data inside each area was subsequently checked comparing the maximum and minimum daily temperature values registered in each station with the corresponding areal average values obtained considering all the stations included in the area. A few stations were so definitively excluded in each zone while two stations were moved to a near area. Eventually, 28 meteorological stations throughout the Italian peninsula were considered, specifically six stations in NW, five stations in NE, seven stations in HP, five stations in CP and five stations in SI.

The spatial distribution of the selected stations, shown in Fig. 4, is quite uniform although a larger number of stations would be desirable.

In our model we have introduced the wind speed information through the so-called Wind-Chill (WC) temperature, computed using the mean wind velocity together with the daily minimum air temperature $T_{d,min}$. Wind Chill calculations have been performed for each sub-area using the following formula [23,24]:

$$WC = 13.12667 + 0.6215T - 13.9468V^{0.16} + 0.4863TV^{0.16}$$

where $T = T_{d,min}$ (°C) and V (m/s) is the mean daily wind speed. This formula applies when the mean wind velocity is greater or equal to 1.3 m/s and the temperature is lower than 10 °C. If these conditions are not satisfied, the Wind Chill value is set equal to the minimum daily air temperature.

Heating Degree Days and Heating Degree Months have then been estimated, by means of the procedure previously described, using the Wind Chill Temperature as well (HDDWC, HDMWC).

The effect of the relative humidity on the electricity demand has been supposed to be relevant in conjunction with warm and hot temperatures only, because the perceived temperature can be higher in such meteorological conditions and, as a consequence, the use of air cooling appliances increases. To take into account these situations in our model, we have introduced the Heat Index variable (HI), also known as 'apparent temperature'. Its definition is the result of extensive biometeorological studies. The Heat Index formula is based upon the lookup table presented by Steadman [25]. The table can be mathematically analyzed to produce the following approximate equation:

$$HI = a + bT + cU + dTU + eTU^2 + fT^2U + gT^2 + hU^2 + iT^2U^2$$

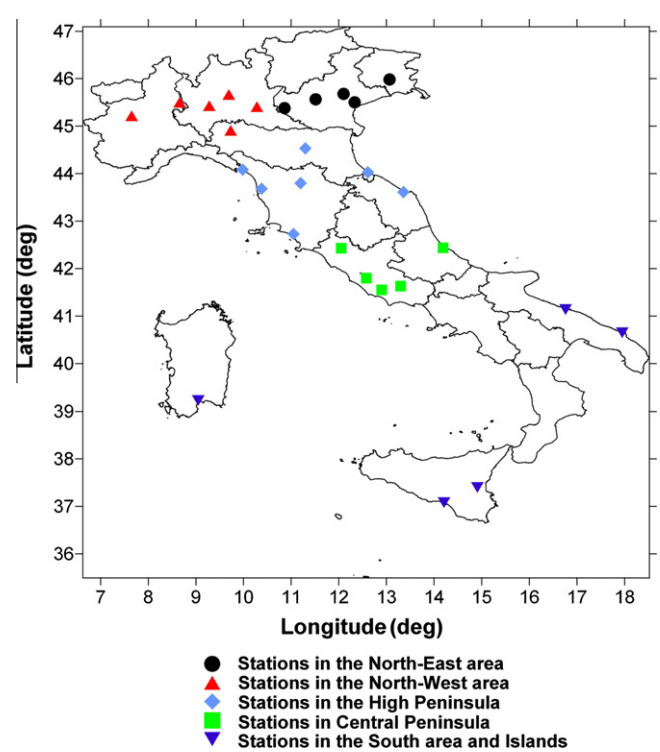

Fig. 4. Localization and grouping of the weather stations on the Italian peninsula (http://clima.meteoam.it/informazioni_en.php?type=generalita).

[1] Climate variability refers to variations in the mean state and other statistics (such as standard deviations and the occurrence of extremes) of the climate on all temporal and spatial scales beyond that of individual weather events.

where T (°C) is the ambient dry bulb temperature, U is the relative humidity (in percentage) and $a,\ldots,i$ are numerical coefficients.

The Heat Index formula has been applied when the temperature is equal or exceeds 27 °C and, simultaneously, the relative humidity is equal or greater than 40%. Meteorological conditions of this kind typically occur in Italy during the daylight, therefore we have evaluated the Heat Index using the maximum daily temperature $T_{d,max}$ and the minimum daily relative humidity $U_{d,min}$. If the above-mentioned conditions are not satisfied, the Heat Index value is set equal to the maximum daily air temperature.

HI values have been computed for each sub-area and, subsequently, Cooling Degree Days and Cooling Degree Months have been estimated using the Heat Index (CDDHI, CDMHI).

Finally, cloud cover information has been introduced in our model, since an increase in cloud cover during daylight can increase lighting demand. Cloud coverage data are registered at the weather stations every three hours and are given as fraction of obscured sky in 1/8 units. Only daylight coverage values greater than 6/8 were considered, then the daily coverage in each station was assessed on the basis of the number of these events. For each given sub-area, the regional daily cloud cover was evaluated averaging the cover estimates performed at the weather stations included in the area. Monthly national values were eventually computed cumulating the daily regional estimates of the five areas considered.

3. Model description

The main target of this study has been to evaluate the impact of the meteorological variables on the monthly electricity demand in Italy using historical data. A parametric multiple linear regression has been preferred over other approaches because the model is simple to use and allows to investigate the influence of the weather and socio-economical variables on the electricity demand in an easy and controlled manner. The developed model can be used to forecast the electricity demand for the next month ahead.

Our model is based on the following assumptions:

- the demand of the next month is estimated on the basis of the actual power consumption of the present month, which takes into account all the effects (socio-economic, climatic, etc.) that determine the demand level on a long-term scale.
- usual seasonal variations between subsequent months are roughly accounted for by assuming that the ratio of the demand in two contiguous months is equal to the corresponding ratio evaluated in the previous year.
- mid and short-term changes of the calendar and of selected weather variables can be used to refine the initial demand estimate based on the previous assumption.

Moving from the above statements, a rough demand estimate for the next month, named $P0$, is thus computed as:

$$P0_{m,y} = C_{m-1,y} \cdot C_{m,y-1}/C_{m-1,y-1}$$

where C is the actual demand; the subscripts 'm' and 'y' denote, respectively, the current month and year while '$m-1$' and '$y-1$' indicate the previous ones.

An improved estimate $P_{m,y}$ is then computed assuming that the correction to the first guess $P0$ depends only on the changes of the meteorological and socio-economic variables between the considered couples of months. To this purpose we introduce an extra variable ΔV as:

$$\Delta V = (V_{m,y} - V_{m-1,y}) - (V_{m,y-1} - V_{m-1,y-1})$$

where V can be any socio-economic or weather variable. ΔV is introduced in order to take account of the different variations of a given variable in homologous couple of months of contiguous years. ΔV highlights the changes in the considered variable and its value is zero whenever the same change occurs on both years.

The improved demand forecast $P_{m,y}$ is thus obtained by means of the following equation:

$$P_{m,y} = P0_{m,y} \cdot (1 + R_{\mathrm{MLR}})$$

where the correction term R_{MLR} is evaluated by means of a Multiple Linear Regression model in which the dependent variable is:

$$R_{m,y} = (C_{m,y}/P0_{m,y}) - 1$$

while the independent ones are the above defined ΔV weather and socio-economical related variables. The general expression of the MLR utilized in this study is

$$R_{m,y} \approx R_{\mathrm{MLR}} = \alpha_1 \Delta V_1 + \alpha_2 \Delta V_2 + \ldots + \alpha_n \Delta V_n$$

where α_i (for $i = 1,2,\ldots,n$) are the regression coefficients determined using the data in the N years preceding the one to be predicted.

The socio-economical variable used is the monthly weighted calendar variable CalW, while the weather related variables considered are:

- cloud cover (subsequently indicated with Cds),
- Heating and Cooling Degree Months estimated using minimum and maximum daily temperatures (HDM, CDM) or, alternatively, the Wind-Chill and the Heat Index (HDMWC, CDMHI).

To select the best threshold values to compute the heating and Cooling Degree-Days, the monthly electricity demand has been simulated considering many possible combinations of the *hot* and *cold* threshold values. Different sets of demand estimates were obtained using all the possible couples of the degree-month variables, considering or not the cloud cover and varying the length of the regression period. Simulations have been performed spanning the 1994–2009 time interval (see next section). Yearly Mean Absolute Percentage Error (MAPE) between the estimated and observed electricity demand values has been calculated for each couple of thresholds and for every year. Best combinations of the threshold values have been searched so as to minimize the average yearly MAPE for the whole period considered.

In all cases, the average yearly MAPE varied slowly around its minimum for T_{max} and T_{min} threshold values included respectively in the 24–26 °C and in the 2–4 °C intervals.

Eventually, a value equal to 25 °C has been chosen to compute both CDDHI and CDD, while threshold values equal to 4 °C and 3 °C have been used to calculate HDD and HDDWC.

4. Model accuracy

The accuracy of the model has been evaluated considering different possible choices of the weather related variables. Demand estimates were performed over the period 1994–2009 using actual demand data till the month to be predicted and measured meteorological data for all the months considered (in the following discussion we shall call '*simulation*' this kind of demand estimate). To predict the monthly demand in a given year, the regression coefficients were computed using the data of the previous 7 years (this length of time gave the best results). Therefore the period used to evaluate the coefficients was shifted year by year and all the available data since the end of 1986 to 2008 were progressively used to simulate the demand data in the 1994–2009 period.

In Table 1 we quantify the overall accuracy of the model with some of the most effective combinations of the meteorological variables we tested. Accuracies obtained with the first guess $P0$

Table 1
Accuracy of the model for different choices of the regression variables.[a] Basic statistics of the Absolute Percentage Errors (APE) over the period 1994–2009.

	Mean APE	Standard deviation	Max APE
PO	2.04	1.69	8.47
Sim_CalW	1.51	1.39	9.21
Sim_CalW_HDMWC3_CDMHI25	0.87	0.68	3.93
Sim_CalW_HDMWC3_CDMHI25 Cds	0.85	0.65	3.22
Sim_CalW_HDM4_CDM25_Cds	0.86	0.67	3.09
Sim_CalW_HDM4_CDM25	0.89	0.71	3.90

[a] Sim_CalW_HDMWC3_CDMHI25_Cds denotes, for instance, a simulation performed using the weighted calendar variable CalW, Heating Degree Months with Wind Chill threshold equal to 3 °C, Cooling Degree Months with Heat Index threshold equal to 25 °C and Cloud Cover Cds.

and with the calendar variable alone are also included for comparison. The Mean Absolute Percentage Error (MAPE) with the related standard deviation and the maximum value of the absolute percentage error (APE) over the whole period are shown. The basic statistics of the APEs is thus performed on 192 values (16 years × 12 months).The smallest APEs are not shown as they are nearly zero.

Considering the MAPEs we observe that:

- a noticeable advantage always results from the use of the meteorological variables in addition to the calendar one. In fact even the less effective choice of the weather variables (HDM4 and CDM25) determines a reduction of the MAPE from 1.51% to 0.89%.
- accuracies obtained with different choices of the weather-related variables are actually quite similar: slightly better results are achieved by introducing the cloud cover information and using Wind Chill and Heat Index, however, as expected, the driving weather variable is the temperature.

The effect of the weather variables becomes more evident when the errors of estimate for the individual months are considered. In Fig. 5 we show the average values of the APEs evaluated for each month with the same previous combinations of variables. Average yearly values are reported too. The related standard deviations are shown in Fig. 6. It can be seen that according to the month considered a specific combination of weather related variables gives the lowest MAPE but differences are usually small. Anyway the effectiveness of the cloud cover variable can be seen in February, March and April, while the improved accuracy for July is very likely due to the apparent temperature effect, accounted for by the CDMHI25 variable.

As the overall accuracy of the model is quite similar with the four different combinations of the weather related variables, we show a detailed comparison between the estimated and actual monthly demand data over the period 1994–2009 for the following set of variables only: CalW, Cds, HDM4 and CDM25. We choose the variables HDM and CDM, rather than the composite variables HDMWC3 and CDMHI25, because humidity and wind velocity data are often less available than temperature alone. Relative percentage errors between the estimated and actual monthly demand data over the 1994–2009 period are plotted in Fig. 7. Errors corresponding to the first guess $P0$ are also shown for comparison. The improvement of the final estimate compared to the first guess is remarkable and the Relative Percentage Error (RPE) is usually within ±2%. The MAPE is 0.86 ± 0.67% (Table 1).

A comparison between the actual and predicted monthly demand is presented in detail for the years 2003 and 2008 (Figs. 8 and 9), which are both peculiar in some respect. In 2003 an anomalous heat event was registered in Italy during the summer months (from the beginning of May to the end of August), while in 2008 a significant downturn of the Italian economy started in the third quarter of the year. The 'zero-level' demand estimates $P0$ are also plotted, so as to show the correction performed by the linear regression model.

Considering the 2003 results, it can be seen a very close correspondence between the actual and predicted values. The absolute percentage error of the monthly estimates ranges between 0.03 (in July) and 1.74 (in September), with a mean value of 0.75 ± 0.55%. In this case the MLR has been applied to years from 1996 to 2002: the retrieved variable coefficients, as well as some indicators that describe their statistical significance are listed in Table 2. As can be seen, all coefficients are statistically significant. The monthly percentage contributions of each model component ΔV_i to the estimated demand P_m, evaluated as $P0_m \cdot \alpha_i \cdot \Delta V_i \cdot 100/P_m$, are listed in Table 3. The effect of the anomalous temperatures is evident on the ΔCDM25 variable in August (4.95%) and September (−7.56%); a relevant correction to the rough estimate $P0$ is also due to the ΔHDM4 contribute in February (3.66%).

The anomalous weather conditions that occurred in 2003, affects the estimates performed by the model in June, July and

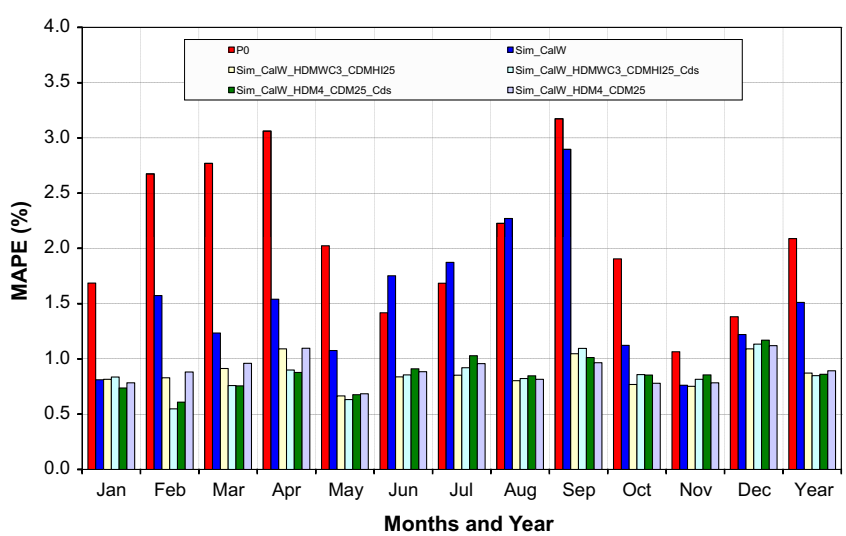

Fig. 5. Model accuracy: monthly and yearly MAPEs obtained with different combinations of the calendar and meteorological variables over the period 1994–2009.

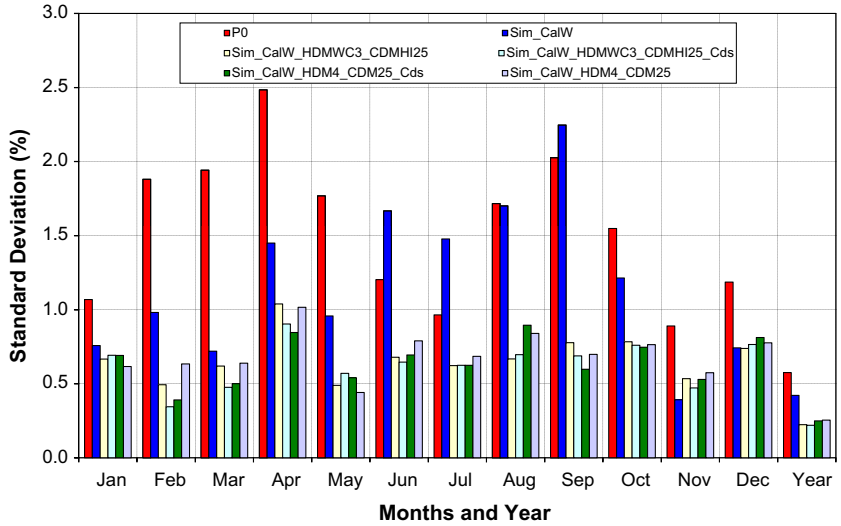

Fig. 6. Model accuracy: standard deviations corresponding to MAPE values reported in Fig. 5.

August of the following year: APEs equal to 3%, 2.19% and 2.84% are actually observed in these months, while the errors are always below 1% elsewhere (see Fig. 7b).

About the result for the year 2008, Tables 4 and 5 show, respectively, the outcome of the MLR and the relative contribution of each model component to the predicted demand. In this case we

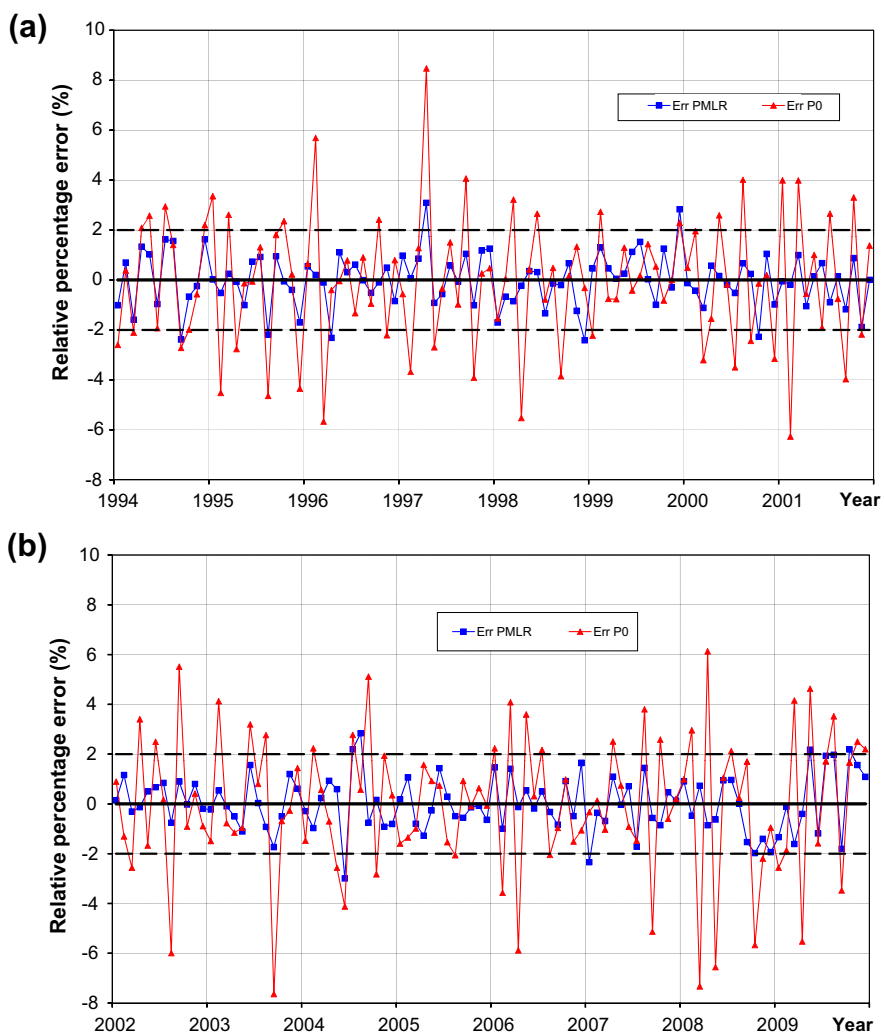

Fig. 7. Model accuracy: relative percentage errors of the monthly demand estimates with respect to the actual data over the periods (a) 1994–2001 and (b) 2002–2009. Model estimates PMLR were obtained with the following variables: CalW, HDM4, CDM25 and Cds. Errors corresponding to the first guess *P*0 are also plotted for comparison.

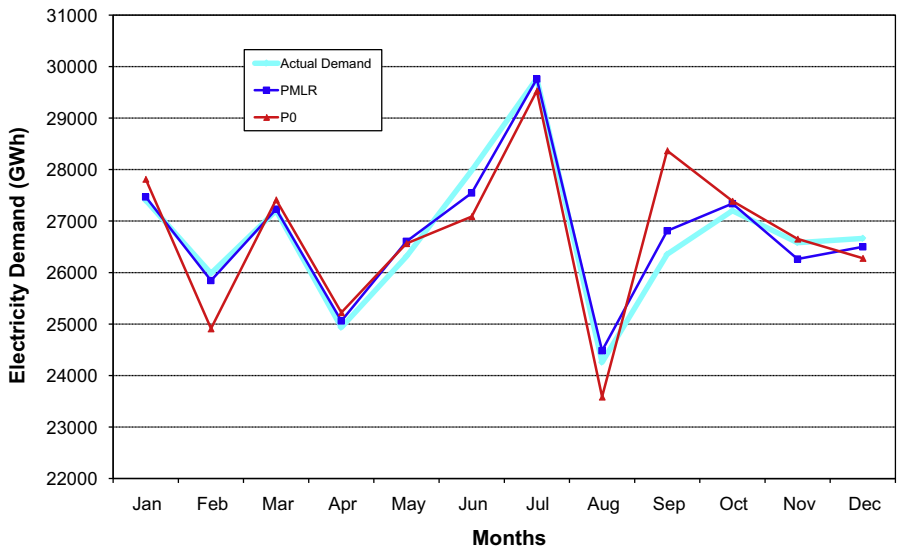

Fig. 8. Comparison between the actual and estimated monthly demand in 2003. Model results were obtained using the following variables: CalW, HDM4, CDM25 and Cds. First guess *P*0 is also plotted for comparison.

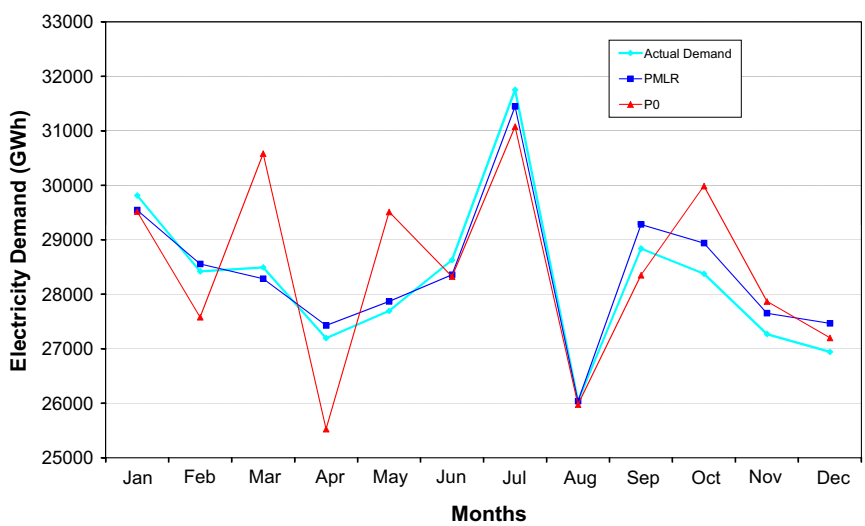

Fig. 9. Comparison between the actual and estimated monthly demand in 2008. Model results were obtained using the following variables: CalW, HDM4, CDM25 and Cds. First guess *P*0 is also plotted for comparison.

observe a remarkable correction of the *P*0 estimate due to the calendar variable, mainly in the months from February to May (consistently with the fact that 2008 was a leap year and that Easter was in April in 2007 while occurred in March in 2008). From September to December we get absolute percentage errors between 1.41% and 1.98%, which are larger than those obtained in the previous 6 months, characterized by a MAPE equal to 0.69%. This is likely to be related to the strong economic downturn in late 2008 which caused a change in the demand trend with respect to the same period in 2007. A more evident effect of the economic turbulence over the 2008–2009 period is actually observed in 2009 (see Fig. 7b), as absolute percentage errors around 2% are obtained in 5 months.

It is worth noting that the calendar variable used in this study has determined a substantial improvement of the demand estimates with respect to other types of calendar description that we previously tested. Particularly, a significant reduction in the monthly absolute percentage errors has been observed in those months which include Christmas and Easter holidays.

Table 2
MLR over the period 1996–2002: estimated variable coefficients and statistical significance indicators.

Variable	Coefficient	Standard error	*t*-Ratio	*p*-Value
ΔCalW	0.03713	0.00192	19.32661	0.00000
ΔCds	0.00010	0.00003	3.46920	0.00084
ΔCDM25	0.00009	0.00001	12.79258	0.00000
ΔHDM4	0.00004	0.00001	5.75787	0.00000

With regard to the contribution of each model component to the monthly demand estimates, in Table 6 we show the largest values of the monthly percentage contributions (absolute values) over the whole period (1994–2009). Calendar contributions are considerable from January to May, consistently with the occurrence of some public holidays (mainly Easter) and leap years. On the other

Table 3
Monthly percentage contribution of each model component to the estimated demand in 2003.

Month	ΔCalW	ΔCDM25	ΔHDM4	ΔCds
January	−1.62	0.00	0.95	−0.59
February	0.91	0.00	3.66	−0.96
March	0.86	−0.02	−1.89	0.36
April	−0.61	0.02	−0.11	0.05
May	−0.74	1.19	−0.14	−0.16
June	0.04	1.59	0.00	0.04
July	0.78	0.01	0.00	−0.02
August	−1.22	4.95	0.00	−0.07
September	1.52	−7.56	0.00	0.25
October	−0.19	−0.62	0.13	0.50
November	−0.65	−0.19	−0.16	−0.48
December	0.27	0.00	1.04	−0.47

Table 4
MLR over the period 2001–2007: estimated variable coefficients and statistical significance indicators.

Variable	Coefficient	Standard error	t-Ratio	p-Value
ΔCalW	0.03444	0.00229	15.03313	0.00000
ΔCds	0.00007	0.00003	2.08651	0.04012
ΔCDM25	0.00010	0.00001	17.56667	0.00000
ΔHDM4	0.00004	0.00001	5.44058	0.00000

Table 5
Monthly percentage contribution of each model component to the estimated demand in 2008.

Month	ΔCalW	ΔCDM25	ΔHDM4	ΔCds
January	0.70	0.00	−0.76	0.15
February	3.06	0.00	0.69	−0.33
March	−7.69	0.00	−0.71	0.28
April	6.83	−0.16	−0.14	0.40
May	−4.46	−1.12	−0.04	−0.27
June	−0.67	0.87	0.00	−0.09
July	2.38	−1.17	0.00	−0.03
August	−2.76	3.29	0.00	−0.29
September	2.91	−0.13	0.00	0.40
October	−1.31	−1.95	−0.06	−0.30
November	−1.13	0.26	−0.14	0.23
December	1.20	0.00	−0.41	0.18

Table 6
Largest values of the monthly percentage contributions of each model component (absolute values) over the period 1994–2009.

Month	ΔCalW	ΔCDM25	ΔHDM4	ΔCds
January	4.02	0.00	2.45	1.00
February	5.85	0.00	3.66	1.40
March	7.69	0.02	2.57	1.07
April	6.83	0.17	1.80	0.91
May	4.46	1.97	0.47	0.72
June	2.21	3.36	0.00	1.06
July	2.78	5.10	0.00	0.47
August	2.76	4.94	0.00	0.48
September	2.91	7.56	0.00	1.00
October	2.52	2.46	0.26	0.91
November	2.39	0.59	1.33	0.66
December	2.56	0.00	2.28	0.81

hand, as expected, the largest contributions of the Cooling and Heating Degree Months occur, respectively, during the summer and winter months. The influence of the cloud cover variable is always below 1% throughout the year.

Table 7
Forecasting accuracy of the model for two different choices of the regression variables. Basic statistics of the Absolute Percentage Errors (APE) over the period 1994–2009.

	Mean APE	Standard deviation	Max APE
For_11_CalW_HDMWC3_CDMHI25_Cds	1.30	1.20	7.83
For_11_CalW_HDM4_CDM25_Cds	1.31	1.19	7.30

5. Forecasting results

In this section we give an example of the applicability of our methodology to perform a medium term forecast (1 month ahead).

The forecasting capability of the model has been evaluated over the period 1994–2009, performing demand estimates for 1 month ahead. Actually a forecast differs from a '*simulation*' because of the use of predicted meteorological values for the next month. The forecasting accuracy is thus determined both by the model and by the weather forecasts accuracies. In this study the estimate of each weather related variable for the month to be predicted was obtained averaging the monthly values observed in the previous years for the month under consideration (a period of 11 years resulted a good choice).

We quantify the overall forecasting accuracy considering the basic statistics of the APE values over the whole period. The results are shown in Table 7, for the two combinations of variables that gave slightly better results as regards the model accuracy. In both cases, MAPEs are around 1.30% and the largest APE is between 7.3% and 7.8%. This worsening is due to the poor estimate of the monthly weather variables: actually the largest APE value is observed during summer 2003 when the anomalous heat wave occurred.

In Fig. 10 we show the average values of the APEs evaluated for each month with the same previous combinations of variables. Average yearly values are reported too. Related model accuracy results are plotted for comparison and it can be seen that the worsening mainly occurs in the summer months from June to September. Forecast MAPE values that are observed in these months range between 1.6% and 2.2% and are about twice those obtained as model accuracy.

6. Conclusions

The main target of this study was to evaluate the influence of the meteorological variables on the Italian monthly electricity demand. A relatively simple multiple regression forecasting model including both weather variables (temperature, relative humidity, wind velocity and cloudiness) and socio-economic factors (calendar variables) has been presented. The non-linear dependence of the electricity demand on temperature has been dealt with using the Cooling and Heating Degree Month variables; the possible effect of wind speed and relative humidity has been taken into account in conjunction with the air temperature conditions through the Wind Chill and the Heat Index Temperatures. A proper calendar variable has been introduced so as to weight each month not simply counting the number of holidays and working days, but specifically taking into account which days of the week are included and considering in detail the various kind of holidays as well as their effect on previous and following days.

The overall accuracy of the model has been evaluated considering different combinations of the weather-related variables and comparing the estimates against actual data over the period 1994–2009. Accuracies obtained are actually quite similar and range between 0.85% and 0.89%. Slightly better results are achieved

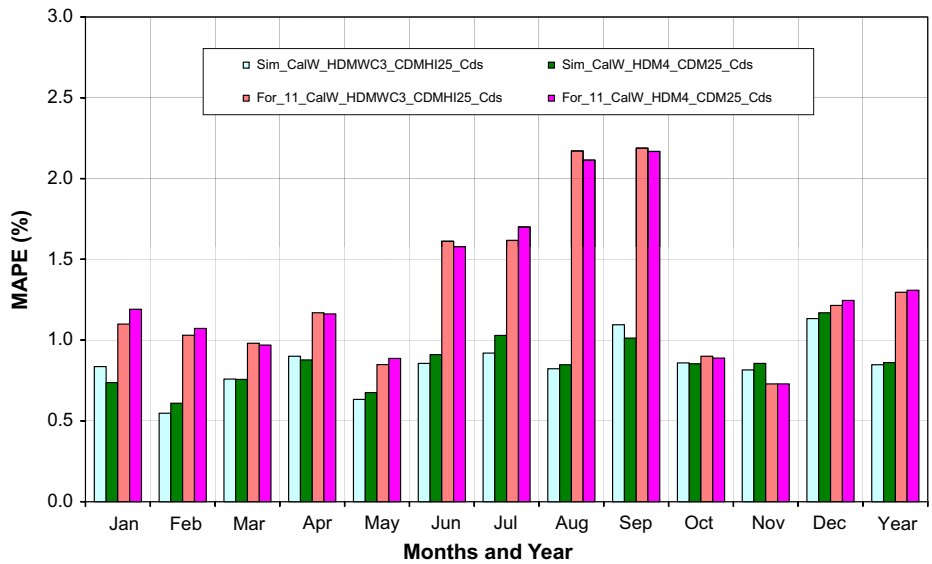

Fig. 10. Forecasting accuracy: monthly and yearly MAPEs obtained with two different choices of the meteorological variables over the period 1994–2009. Related model accuracy results are also plotted for comparison.

by introducing the cloud cover information and using Wind Chill and Heat Index, however, as expected, the driving weather variable is the temperature. The effect of the meteorological variables is more evident when the errors of estimate for the individual months are considered. According to the month considered, a specific combination of the weather related variables can give the lowest MAPE but differences are usually small.

The proposed model clearly identifies the influence of the weather conditions on the aggregated national electricity demand.

Considering the largest percentage contribution (absolute value) of the model components to the monthly demand estimates, it has been shown that: the role of the calendar component is considerable from January to May with weights between 4.0% and 7.7%; the contribution of the Cooling Degree Months ranges from 3.4% to 7.6% during the summer months while that of the Heating Degree Months is between 2.3% and 3.7% during the winter ones. The influence of the cloud cover variable is always below 1% throughout the year.

The model has also been used to predict the electricity demand for the next month ahead. An overall forecasting accuracy around 1.3% (MAPE) over the whole period (1994–2009) has been found. The worst MAPEs have occurred from June to September with values between 1.6% and 2.2%, which are about twice those obtained as model accuracy in the same months.

Of course an increase of the accuracy of the weather forecast will improve the accuracy of the monthly electricity demand forecast.

We want to remark the good accuracy of the electricity demand simulations and forecasts obtained during summertime, despite the statistically evident increase in the last decades of both the heat waves duration and the daily maximum temperatures in Italy. These summer performances of the model could be very useful in the future when, according to some model results [26], heat waves in Europe and North America are likely to become more intense, more frequent and longer lasting in the second half of the 21st century.

The proposed methodology could be directly applicable to demand data disaggregated by sectors (industrial, residential, etc.) and geographically. Moreover, in our opinion, it could be easily adapted to perform forecasts at shorter time scales (daily for instance).

Acknowledgements

This work has been financed by the Research Fund for the Italian Electrical System under the Contract Agreement between RSE (formerly known as ERSE) and the Ministry of Economic Development – General Directorate for Nuclear Energy, Renewable Energy and Energy Efficiency stipulated on July 29, 2009 in compliance with the Decree of March 19, 2009.

The authors are thankful to Tiziano Colombo of the CNMCA and Barbara Santini, Andrea Lupi and Claudio La Ianca of the Terna S.p.A. – Regulatory Affairs Department for the expression of interest in our research activity and for supplying, respectively, meteorological and electricity demand data.

Finally the authors also thank the anonymous referees for their useful comments, suggestions and criticism.

References

[1] IPCC. Climate change 2007: the physical science basis. In: Solomon S, Qin D, Manning M, Chen Z, Marquis M, Averyt KB, et al., editors. Contribution of working group I to the fourth assessment report of the intergovernmental panel on climate change, vol. 5. Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press; 2007.

[2] Hor C-L, Watson SJ, Majithia S. Analyzing the impact of weather variables on monthly electricity demand. IEEE Trans Power Syst 2005;20(4):2078–85.

[3] Hor C-L, Watson SJ, Majithia S. Daily load forecasting and maximum demand estimation using ARIMA and GARCH. In: 9th International conference on probabilistic methods applied to power systems. KTH, Stockholm, Sweden – June 11–15, 2006.

[4] Amato AD, Ruth M, Kirshen P, Horwitz J. Regional demand responses to climate change: methodology and application to the commonwealth of Massachusetts. Clim Change 2005;71:175–201.

[5] Feinberg EA, Genethliou D. Load forecasting. In: Chow JH, Wu FF, Momoh JJ, et al., editors. Applied mathematics for restructured electric power systems: optimization, control, and computational intelligence. New York: Springer; 2005. p. 269–85.

[6] Widén J, Wäckelgård E. A high-resolution stochastic model of domestic activity patterns and electricity demand. Appl Energy 2010;87:1880–92.

[7] Vassileva I, Wallin F, Dahlquist E. Analytical comparison between electricity consumption and behavioral characteristics of Swedish households in rented apartments. Appl Energy 2012;90:182–8.

[8] Chung M, Park HC. Building energy demand patterns for department stores in Korea. Appl Energy 2012;90:241–9.

[9] Lam JC, Tang HL, Li DHW. Seasonal variations in residential and commercial sector electricity consumption in Hong Kong. Energy 2008;33:513–23.

[10] Kyriakides E, Polycarpou M. Short term electric load forecasting: a tutorial. In: Chen K, Wang L, editors. Trends in neural computation studies in computational intelligence, vol. 35. Springer-Verlag; 2007. p. 91–418.

[11] Hahn H, Meyer-Nieberg S, Pickl S. Electric load forecasting methods: tools for decision making. Eur J Oper Res 2009;199:902–7.
[12] Alfares HK, Nazeeruddin M. Electric load forecasting: literature survey and classification of methods. Int J Syst Sci 2002;33(1):23–34.
[13] Zhu S, Wang JZ, Zhao W, Wang J. A seasonal hybrid procedure for electricity demand forecasting in China. Appl Energy 2011;88:3807–15.
[14] Mirasgedis S, Sarafidis Y, Georgopoulou E, Lalas DP, Moschovits M, Karagiannis F, et al. Models for mid-term electricity demand forecasting incorporating weather influences. Energy 2006;31:208–27.
[15] Moral-Carcedo J, Vicéns-Otero J. Modelling the non-linear response of Spanish electricity demand to temperature variations. Energy Econ 2005;27: 477–94.
[16] Trenberth KE, Jones PD, Ambenje P, Bojariu R, Easterling D, Klein Tank A, et al. Observations: surface and atmospheric climate change. In: Solomon S, Qin D, Manning M, Chen Z, Marquis M, Averyt KB, et al., editors. Climate change 2007: the physical science basis. Contribution of working group I to the fourth assessment report of the intergovernmental panel on climate change. Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press; 2007. p. 311–2.
[17] Grazzini F, Ferranti L, Lalaurette F, Vitart F. The exceptional warm anomalies of summer 2003. ECMWF Newslett 2003;99:2–8.
[18] Schär C, Vidale PL, Lüthl D, Frel C, Häberli C, Liniger MA, et al. The role of increasing temperature variability in European summer heatwaves. Nature 2004;427:332–6.
[19] Valor E, Meneu V, Caselles V. Daily air temperature and electricity load in Spain. J Appl Meteorol 2001;40:1413–21.
[20] Bessec M, Fouquau J. The non-linear link between electricity consumption and temperature in Europe: a threshold panel approach. Energy Econ 2008;30:2705–21.
[21] Giannakopoulos C, Psiloglou BE. Trends in energy load demand for Athens, Greece: weather and non-weather related factors. Clim Res 2006;31:97–108.
[22] Pardo A, Meneu V, Valor E. Temperature and seasonality influences on Spanish electricity load. Energy Econ 2002;24:55–70.
[23] Media Guide to NWS Products and Services, National Weather Service Forecast Office, Monterey, CA; 1995.
[24] New Wind Chill Temperature Index, Office of Climate, Water and Weather Services, Washington, DC; 2001.
[25] Steadman RG. The assessment of sultriness; Part I: a temperature-humidity index based on human physiology and clothing science. J Appl Meteorol 1979;18:861–73.
[26] Meehl GA, Tebaldi C. More intense, more frequent and longer lasting heat waves in the 21st century. Science 2004;305(5686):994–7.